Estimation of Mineral Abundance From Hyperspectral Data Using a New Supervised Neighbor-Band Ratio Unmixing Approach

Kevin Siebels, Kalifa Goïta, and Mickaël Germain

***Abstract*— This article compares the ability of nine unmixing models, including radiative transfer (RT) models as well as a new nonlinear unmixing approach called neighbor-band ratio unmixing (NBRU), to obtain mineralogical information from hyperspectral data. Their performance in estimating mineral abundances of 94 crafted mineral mixtures was first assessed. NBRU led to the best results among non-RT models with mean and median errors of 9.8% and 7.4%, respectively. Hapke's and Shkuratov's RT models obtained 6.5% and 5.6%, and 6.7% and 4.7%, respectively. In a second experiment, the mapping ability of six non-RT models and their robustness when facing endmember variability were evaluated. The assessment was performed on an AVIRIS hyperspectral image of the widely studied Cuprite area, NV, USA. Comparisons with validation maps showed that NBRU retrieved the best spatial distributions for seven of the nine minerals mapped.**

***Index Terms*— Hyperspectral, mapping, minerals, nonlinear, shape based, spectral unmixing.**

I. Introduction

THE assessment of surface conditions on terrestrial planets and solar rocky bodies almost entirely relies on remote sensed data due to the limited access to groundtruths. However, surface characterization requires dealing with subpixel targets. These objects contribute partially to the reflectance spectrum of the instantaneous field of view (the pixel) and appropriate methods are therefore needed for their identification and quantification [1]–[3].

Spectral mixture analysis (SMA) has been developed to provide answers to these problems. SMA approaches typically model a reflectance spectrum by mixing the spectral signatures of the pure components involved, weighted by their surface abundance [1]–[3]. Unmixing thus consists in inverting the SMA by calculating the material abundances that best match the model with the measured reflectance spectrum.

The most widely used SMA approach when retrieving relative abundances from a measured reflectance spectrum is based on the linear mixing model (LMM) [1], [4].

The central hypothesis of LMM is that a reflectance spectrum is a linear combination where spectra of pure materials (called *endmembers*) contribute proportionally to their abundance (1). This assumption is valid when materials lie in spatially segregated areas [5]. In such a situation, incident light rays interact with a single material. Consequently, the reflected part conserves the proportion of the materials in the instantaneous field of view and conveys this information toward the sensor

$$y_i = \sum_{r=1}^{N} (a_r * e_{r,i}) + \eta \tag{1}$$

where y_i is the ith spectral band of the modeled reflectance spectrum y, i ranging from one to the total number of spectral bands B. N is the number of endmembers involved, $e_{r,i}$ is the ith spectral band of the rth endmember, and a_r is the rth abundance associated with endmember e_r. Parameter η stands for the residual noise and modeling error. To have a fully physical meaning, the calculated abundances must follow nonnegativity (ANC) and the sum-to-one constraints (ASC)

$$\begin{cases} a_r \geq 0 \forall r \\ \sum_{r=1}^{N} a_r = 1. \end{cases} \tag{2}$$

The abovementioned approach assumes identical spectral signatures for both reference and observed materials. However, illumination changes, shade, and bidirectional reflectance distribution function (BRDF)-related effects may cause endmember variabilities resulting in deviations from optimal unmixing performances [6], [7]. Because these effects primarily affect brightness, common amplitude-based criteria that assess the fit between modeled and measured spectra are particularly sensitive [8], [9].

In order to cope with these brightness variabilities, other alternatives have been proposed to assess the spectral fitting. Chen *et al.* [8] employed sequential quadratic programming

Manuscript received October 18, 2019; revised December 24, 2019; accepted January 15, 2020. Date of publication March 26, 2020; date of current version September 25, 2020. This work was supported in part by the Fonds de recherche du Québec—Nature et Technologie (FRQNT)—Programme de recherche en partenariat sur le développement durable du secteur minier under Grant 2015-MI-193001 and in part by the National Science and Engineering Research Council (NSERC) of Canada—Discovery Program under Grant RGPIN-2018-06101. *(Corresponding author: Kevin Siebels.)*

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Digital Object Identifier 10.1109/TGRS.2020.2969577

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to maximize the fit using four alternative metrics: Spectral Angle Mapper (SAM) [10], cross correlogram spectral matching (CCSM) [11], spectral information divergence [12], and Euclidian distance (ED). Comparisons when assessing abundances of bar sand, lichen, and dry vegetation from hyperspectral images showed that shape-based criteria systematically outperformed the ED criterion. Tits *et al.* [9] highlighted the advantage of shape-based criteria when unmixing vegetation species with large variations in illumination.

Besides the shape-based metrics of reflectance spectra, other approaches apply a transformation prior to the unmixing process. For instance, Somers *et al.* [13] used selected bands of reflectance spectra as well as their first and second derivatives in an integrated spectral unmixing (InSU) process. The authors showed that the addition of the shape-related information enhances the unmixing performances in an orchard context. The continuum removal introduced in [14] has proven to help in mapping studies as well [15]–[17] while wavelets have been proposed for dimensionality reduction and feature extraction in spectral unmixing and classification perspectives [18]–[20].

Previous methods assume a linear relationship between a material's abundance and its spectral contribution. However, when incident radiation undergoes multiscattering before reaching the sensor, the linear hypothesis can no longer be assumed [5], [21]. Particulate surfaces are typical situations where nonlinear relationships are encountered. Planetary regoliths, beaches, and sand deserts are composed of particles, where the scale of the mixture is smaller than the path length of the photons inside the material [5], [22], [23]. These so-called *intimate mixtures* lead to complex nonlinear relationships depending on numerous factors: size, relative size, and shape of the particles [5], [24]–[29], material packing [24], [25], angles of illumination [22], [29], etc. A widely observed nonlinear relationship is the higher influence of dark materials relative to their abundance [28], [30], [31].

A well-established method to model the reflectance spectra of mineral mixtures in the laboratory is the radiative transfer (RT) model introduced by Hapke [32]. The inversion of Hapke's model with well-constrained particle sizes achieves estimates of common mineral abundances with less than 10% error [21], [31]. Although reasonable approximations can be made [31]–[33], some parameters of Hapke's model are unavailable outside laboratories or without extensive fieldwork.

Consequently, poor performances of linear approaches in retrieving mineral abundances and difficulties in inverting RT models gave rise to many nonlinear models. Several of these approaches consider the multiscattering phenomenon as successive additions of linear combinations. Although bilinear models assume scattering of order higher than two negligible, others integrate more interactions in their equations. Comprehensive overviews of nonlinear models can be found in [34] and [35].

Model performances in mapping mineral species are frequently validated by visual comparison with other established models or according to their spectral reconstruction errors. Also, their ability to estimate endmember abundances are frequently assessed on simulated spectra.

In this article, we introduce a new unmixing approach built on shape-based reasoning. The inversion of this new approach is as straightforward as the LMM and aims to capture the nonlinear relationship between the mineral abundances and their spectral contribution.

The ability of the proposed approach in retrieving mineral abundances is compared with eight other models. Quantitative performances are assessed by unmixing the spectra of 94 crafted mineral mixtures from the NASA Reflectance Experiment Laboratory (RELAB). Among the existing models considered, five were also evaluated in mapping mineral distributions by unmixing the AVIRIS hyperspectral image of Cuprite, NV, USA. The abundance maps were compared with published maps. In both experiments and preprocessing steps.

A. Existing Models Considered

Bilinear models propose to add secondary scattering contributions to the linearly reconstructed spectrum to induce nonlinear behavior. Interactions of order higher than two are not modeled. Bilinear models can be generalized by the following equation:

$$y_i = \sum_{r=1}^{N} (a_r \times e_{r,i}) + \sum_{j=1}^{N_1} \sum_{k=1}^{N_2} \beta_{j,k} e_{j,i} \odot e_{k,i} + \eta \qquad (3)$$

where y_i is the ith reflectance value of the modeled spectrum, N is the number of endmembers involved in the first-order interactions, N_1 and N_2 are the number of endmembers involved in the second-order interactions. Parameter a_r is the abundance of the rth material, $e_{r,i}$ is the ith reflectance value of the rth endmember, $\beta_{j,k}$ is a scalar quantifying the contribution of the Hadamard (element-wise) product of the jth, and kth endmembers $e_{j,i} \odot e_{k,i}$, and η combines the noise and modeling errors. Models encompassed by the general equation (3) reduce to the LMM (1) when β parameters are all equal to zero.

Fan *et al.* [36] assume that bilinear contributions are entirely quantified by the abundance of the materials involved. The assumption underlying this hypothesis is that the probability for illumination rays to interact with a material is proportional to its abundance. The model proposed by Fan and coauthors (FM) defines second-order interaction coefficients as the product of the abundances of the interacting materials, $\beta_{jk} = a_j a_k$. Both ANC and ASC (2) are imposed on abundance parameters and self-interactions are not considered. The model can be expressed by the following equation:

$$y_i = \sum_{r=1}^{N} (a_r \times e_{r,i}) + \sum_{j=1}^{N} \sum_{k=j+1}^{N-1} a_j a_k e_{j,i} \odot e_{k,i} + \eta. \qquad (4)$$

Although being intuitive, strictly relating bilinear interactions with the LMM part of the model is a particularly restrictive constraint that can be too rigid when modeling complex spectral behaviors.

Halimi *et al.* [37] refer to [38] to add a coefficient β to circumvent the restrictive behavior of FM. In this generalized bilinear model (GBM), the coefficients $\beta_{j,k}$ quantify

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the amount of second-order interaction between jth and kth endmembers. An important property of GBM is to unify both LMM ($\beta = 0$) when no second-order interaction is assumed and FM ($\beta = 1$) when the contribution of second-order scattering is fully defined by materials abundance

$$y_i = \sum_{r=1}^{N} a_r \times e_{r,i} + \sum_{j=1}^{N-1} \sum_{k=j+1}^{N} \beta_{j,k} a_j a_k e_{j,i} \odot e_{k,i} + \eta \quad (5)$$

$$\begin{cases} a_r \geq 0 & \forall r \\ 1 \geq \beta_{j,k} \geq 0 & \forall j, \forall k \\ \sum_{r=1}^{N} a_r = 1. \end{cases} \quad (6)$$

The authors applied the GBM model on a subset of an AVIRIS image of the Cuprite Hills, NV, USA. The model showed better performances in reconstructing the measured spectra compared with LMM and FM.

Altmann *et al.* [39] proposed a polynomial postnonlinear mixing model (PPNMM) to induce nonlinearities to a spectrum by adding a transformed linear model using a polynomial function of degree 2. The obtained spectrum includes both different-material and same-material interactions. Higher orders were considered negligible and hardly distinguishable from noise. A single parameter β quantifies the bilinear terms. The model can be written

$$y_i = \sum_{r=1}^{N} a_r \times e_{r,i} + \beta \left[\sum_{r=1}^{N} (a_r \times e_{r,i}) \right] \odot \left[\sum_{r=1}^{N} (a_r \times e_{r,i}) \right] + \eta. \quad (7)$$

The PPNMM was compared by authors with LMM, FM, and GBM using synthetic and real hyperspectral images. The four models were used to create their own synthetic data sets with abundances and hyperparameters randomly defined within appropriate constraints. Although each model performed well in unmixing its respective data set, the PPNMM showed better results in retrieving material abundances from spectra mixed by other algorithms. It reached better reconstruction performances on remote sensed data sets as well. No constraint was applied by authors on parameter β.

These bilinear models form the core of several unmixing methods [40]–[43]. They are frequently used to simulate hyperspectral data [41], [44] or as reference methods in comparative experiments [41], [43].

Heylen and Scheunders [45] proposed a model where all multiscattering contributions are considered and weighted by probabilities of interactions. Their model, based on physical reasoning, considers all orders of same-material and different-material interactions. This multilinear mixing model (MLM) can be written as follows:

$$y_i = \frac{(1 - P) \sum_{r=1}^{N} a_r \times w_{r,i}}{1 - P \sum_{r=1}^{N} a_r \times w_{r,i}} + \eta \quad (8)$$

where w_i is the albedo of the ith material, and P is the probability that a light ray undergoes further interactions. Latter parameter quantifies the nonlinear behavior of the model. The model reduces to LMM when $P = 0$. Although the value of P has only a physical meaning between 0 and 1, authors left the parameter unbounded in the inversion process

$$\begin{cases} a_r \geq 0 \forall r \\ \sum_{r=1}^{N} a_r = 1. \end{cases} \quad (9)$$

Authors directly used endmembers retrieved with an endmember extractor algorithm to unmix an AVIRIS image of the Cuprite geological site. However, such a strategy was avoided with laboratory measurements. Instead, the authors included a P-dependent reflectance-albedo relationship in the model. The MLM then becomes [45]

$$y_i = \frac{(1 - P) \sum_{r=1}^{N} \frac{a_r \times e_{r,i}}{P \times e_{r,i} + 1 - P}}{1 - P \sum_{r=1}^{N} \frac{a_r \times e_{r,i}}{P \times e_{r,i} + 1 - P}} + \eta. \quad (10)$$

Although significant errors were observed in retrieving mineral abundances of binary quartz-alunite mineral mixtures, the model showed better performances when compared with LMM, GBM, and PPNMM. Like bilinear models, the MLM forms the basis of several more recent approaches [46]–[49]. Recently, the multiharmonic postnonlinear mixing model (MHPNMM) was proposed by Tang *et al.* [43] to model macroscopic and microscopic (intimate) spectral mixtures encountered in natural scenes. The model is based on the MLM and on the p-harmonic mixture model (pHMM) [50] to combine multiple reflection effects and high-order spectral mixture.

In its closed form, MHPNMM can be defined as [43]

$$y_i = (1 - P) \sum_{r=1}^{N} a_r \times e_{r,i} + (1 - P) \times \frac{P \times b \sum_{i}^{R} a_r [e_{r,i} - \cos(e_{r,i}) - \sin(e_{r,i})]}{1 - P \times b \sum_{i}^{R} a_r [e_{r,i} - \cos(e_{r,i}) - \sin(e_{r,i})]} \quad (11)$$

where b is the parameter scaling the nonlinear effect. The MHPNMM was tested by its authors on simulated spectra generated by the LMM, GBM, PPNMM, and MLM. It showed the best reconstruction performances on AVIRIS image of Cuprite and obtained abundance maps consistent with the other models.

Two RT models are also considered in this article. Hapke's RT model [22], [32] depends on the real and imaginary parts of the optical indexes, mineral internal scattering, porosity, mineral grain sizes, and measurement geometry to fully model reflectance spectra.

However, with reasonable assumptions, major simplifications can be made [31]. When the opposition effect is considered negligible between 15° and 40° [31] and when the particles are assumed to be much larger than the wavelengths measured, of similar shape, and randomly oriented, the mixing process then becomes a linear combination of single scattering albedos (SSA) [22], [32]. The relationship is described by the following equation:

$$y(i, e) = \frac{w}{4(\mu + \mu_o)} [H(\mu) H(\mu_o)] \quad (12)$$

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where w is the SSA of the particulate material, $\mu_0 = \cos(i)$ with i being the incident illumination angle, $\mu = \sin(e)$ with e being the reflection angle. Finally, H is Hapke's approximation of Chandrasekhar's equations [51] describing multiscattering in a particulate medium

$$H(\mu) = \frac{(1+2\mu)}{(1+2\mu\gamma)} \tag{13}$$

with $\gamma = (1-w)^{1/2}$.

The coefficients weighting the SSA in the spectral mixtures depend on the densities, grain size, and mass fractions of the endmembers. This coefficient is classically referred to as the F-parameter and is defined by the following equation [31]:

$$F_n = (M_n/\rho_n d_n)/\sum_r^N (M_r/\rho_r d_r) \tag{14}$$

where F_n is the relative geometric cross section of the nth component and M_n, ρ_n, and d_n are its mass fraction, density, and particle diameter, respectively. This form of the model has been validated in [26], [31], and [52].

In 1999, Shkuratov [53] proposed an optical geometrical RT model to reconstruct the reflectance spectra of particulate surfaces. The model is a function of the real and imaginary parts of refractive indexes, the grain sizes, the effective average optical pathlength, the volume fractions occupied by the particles, and the mineral abundances. Shkuratov's model has been tested and validated on crafted mineral mixtures [28] and applied in remote sensing contexts [54], [55].

II. Neighbor-Band Ratio Unmixing Approach

The transformation of reflectance spectra into derivatives has shown interesting results in unmixing research [13], [56], [57]. Derivatives are, however, independent of the amplitude of the reflectance spectrum and are, therefore, unable to handle the well-known unequal spectral influence of minerals of various brightness. Other transformations have also been introduced to isolate the shape information from reflectance spectra and have shown their robustness in remote sensing contexts as well (see Section I). A method combining this robustness with accurate abundance estimations would be of great benefit.

For the proposed function, let s_i be the reflectance value at the ith spectral band and s_{i+1} the value at the following band. Let Δ be the variable defining the difference between the two former parameters: $\Delta = s_i - s_{i+1}$. Finally, let s_i^* be the newly defined variable resulting from the proposed transformation of s_i: $s_i^* = f(s_i)$ with $f(\cdot)$ being the proposed transformation function.

To reach high robustness and good accuracy in estimating mineral abundances, a transformation should meet the two following conditions:

1) Increase the expression of the spectral features of low amplitude spectra to better match with the nonlinear behavior of spectral mixtures. Mathematically, the derivative $f'(s_i)$ of $f(\cdot)$ at $s_i + \Delta$, is inversely proportional to $s_i + \Delta$.
2) Preserve the absence of spectral feature at s_i when the difference with s_{i+1} is equal to zero ($\Delta = 0$). The slope of f at s_i, expressed by its derivative $f'(s_i)$, must be equal to zero.

Functions with multiplicative-inverse relations as their derivative fulfill these criteria

$$f'(s_i) = \frac{\Delta}{(s_i+\Delta)^n} \quad \text{with } n \geq 1 \tag{15}$$

when $n = 1$, the integration of the derivative becomes

$$\begin{aligned} f(s_i) &= \int f'(s_i) ds_i = \int \frac{\Delta}{s_i+\Delta} ds_i \\ &= \Delta \ln(s_i+\Delta) + K \end{aligned} \tag{16}$$

where K is a real constant. The general equation for $n > 1$ is

$$\begin{aligned} f(s_i) &= \int f'(s_i) dS_i = \int \frac{\Delta}{(s_i+\Delta)^n} ds_i \\ &= K - \frac{\Delta}{(n-1)(s_i+\Delta)^{n-1}}. \end{aligned} \tag{17}$$

In the proposed approach, we considered $n = 2$ to ease the integration and its application. The transformation function then becomes

$$f(s_i) = \int f'(s_i) dS_i = \int \frac{\Delta}{(s_i+\Delta)^2} ds_i = K - \frac{\Delta}{(s_i+\Delta)}. \tag{18}$$

$K = 1$ means that the transformation leads to a value of 1 with a flat reflectance spectrum ($\Delta = 0$)

$$s_i^* = f(s_i) = 1 - \frac{\Delta}{s_i+\Delta} = \frac{s_i}{s_{i+1}}. \tag{19}$$

As shown in (19) the function can be summarized as a ratio of contiguous spectral bands, or neighbor band ratios (NBRs).

The neighbor-band ratio unmixing (NBRU) is thus proposed in this article to deal with this nonlinear relationship between mineral abundances and their spectral contribution as well as to cope with the endmember variability encountered in natural scenes. On the contrary, with respect to derivative-related methods, this transformation expresses shapes in a nonlinear way, depending on the amplitude (the "height") of the reflectance spectrum considered.

An NBR expresses the steepness of a slope relatively to the amplitude of the reflectance spectrum. A value higher than 1 indicates a descending slope, whereas a value lower than 1 indicates an ascending one. The distance from 1 is proportional to the steepness of the slope and inversely proportional to the amplitude of the reflectance spectrum. Consequently, NBR of low amplitude spectra are overexpressed relatively to high amplitude spectra. This is illustrated in Fig. 1.

As shown in Fig. 1, the general logarithmic aspect of the reflectance-to-NBR relationship is similar to the reflectance-to-SSA relation of Hapke's model.

In the proposed NBRU approach, the transformation from reflectance spectra to NBR spectra is applied to the endmembers. The individual contributions of a mineral to the shape of the multimineral spectrum, expressed as NBR, are assumed to be proportional to its abundance. Consequently, a linear combination of the endmembers in their NBR form

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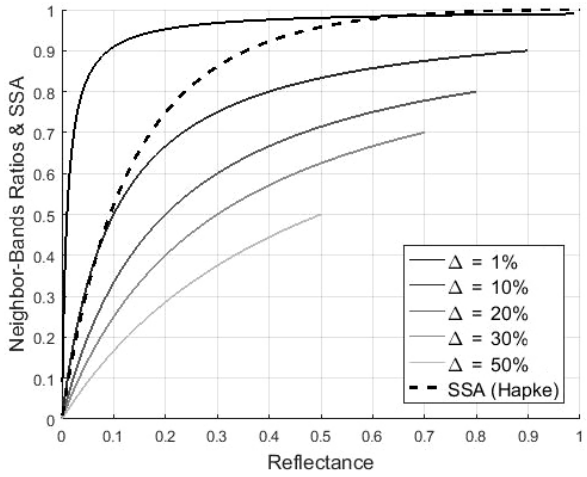


Fig. 1. Hapke's SSA (dashed line) and NBR (s_i^*) for five different Δ (solid lines).

is performed. The approach thus expresses the shape of the endmembers in a nonlinear way

$$y_i^* = \sum_{r=1}^{N} e_i^* \times a_r + \eta^* = \sum_{r=1}^{N} \frac{e_i}{e_{i+1}} \times a_r + \eta^* \qquad (20)$$

where $y*$ is the modeled spectrum in its NBR form and e_r^* is the rth endmember in its NBR form. Parameter η^* is the noise and model error in NBR. Abundance parameters a_r respect both ANC and ASC.

Fig. 2 illustrates how several pairs of reflectance spectra with 50%/50% abundance mix in a nonlinear way when using NBR. The five spectra mixed 2 × 2 are composed of two contiguous spectral bands with a Δ of 5%. Their first band s_i is of 1%, 21%, 41%, 61%, and 81% and their second band is of 6%, 26%, 46%, 66%, and 86%, respectively. The figure shows the projection of the five endmembers from the reflectance (black disks) to the NBR space, their mixing with 50%/50% abundances (white diamond), and finally, the projection of the result from the NBR space back to the reflectance space (black diamonds). The results of classic LMM are also illustrated for comparison (white disks).

Fig. 2 shows how dark materials have a higher spectral influence relative to their abundance when mixed with NBR. This is illustrated by the gaps between the NBR mixing (black diamonds) and the LMM mixing (white disks) on the reflectance axis. The mixing is highly nonlinear with darkest materials due to the relationship between absolute reflectance and NBR, as illustrated by the thick black curve. This trend is in keeping with the overwhelming influence of darkest minerals in mixtures (see Section I).

The proposed NBRU approach aims to handle two problems frequently encountered when unmixing spectra of natural geological scenes. At first, we expect its shape-based reasoning gives higher robustness when facing endmember variabilities. Second, because of the similarity between reflectance-to-NBR and reflectance-to-SSA relationships, the proposed nonlinear relation is expected to cope well

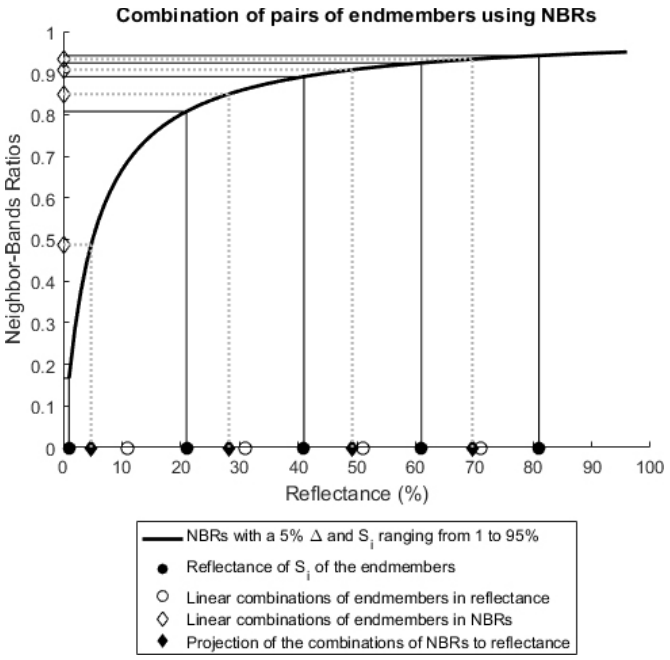


Fig. 2. Comparison between linear combinations of reflectance and NBR. The five endmembers are made up of two spectral bands. The distance between the two bands (Δ) is of 5% and their first (lowest) band s_i is 1%, 21%, 41%, 61%, and 81% of reflectance.

with the unequal spectral contributions of minerals of various brightnesses.

III. Experimental Data

NBRU and eight existing methods were tested on spectra of crafted mineral mixtures from the NASA RELAB at Brown University, publicly available at www.planetary.brown.edu/relab/. The data set used hereunder includes 20 spectra of pure minerals as well as 94 spectra of mixtures made of two, three, or four minerals with known mass fractions. Specific gravities from [58] and [59] were used to convert the volume abundances estimated to mass fractions.

Because this experiment strictly intends to assess the ability of the models in reproducing the spectral phenomenon occurring in intimate mineral mixtures, only the endmembers present (two, three, or four endmembers, depending on the mixture) were considered in the unmixing. Spectral preprocesses were also avoided since these operations could eventually blur the results and complexify their interpretation. While model performances could be enhanced with these methods, exploring the variety of preprocessing and endmember selection algorithms is thus out of the scope of this research.

NBRU and five other methods were also tested on a subset of the AVIRIS image of the widely studied Cuprite area, NV, USA. The AVIRIS sensor is an airborne whiskbroom scanner from NASA Jet Propulsion Laboratory that acquires cross-track images. The hyperspectral image is made of 224 spectral bands ranging from 370 to 2500 nm and with a bandwidth of approximately 10 nm [60]. The data set was acquired in

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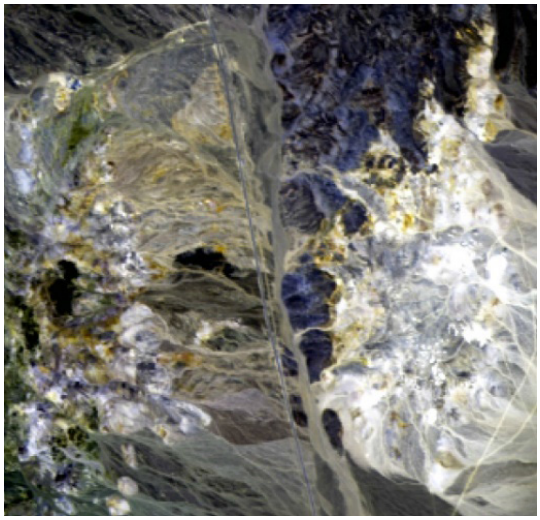


Fig. 3. Color-enhanced RGB subset of the AVIRIS Cuprite data set.

June 1997 and is freely available in reflectance at the official AVIRIS website, www.aviris.jpl.nasa.gov. The subset is of dimensions 300 × 300 pixels and encompasses an area with important mineralogical variability (Fig. 3).

The area is particularly suited to assess performances of unmixing algorithms with geological scenes. Indeed, the absence of vegetation, dry climate, and good outcrop exposure give ideal conditions to measure reflectance spectra with little noise or undesirable interference. Additionally, a large variety of alteration and Fe-bearing minerals provide spectra with strong key features. The area is characterized by two well-exposed zones of strong hydrothermal alterations [61], [62].

In this experiment on a real AVIRIS hyperspectral image of the Cuprite area, 14 endmembers were used in the unmixing. All of them were selected from the USGS spectral library [63]. Considering the lack of *a priori* knowledge on mineral grain size and chemical composition, we selected, for each mineral, the endmember of higher mean correlation with the spectra of the AVIRIS image. Correlation is used as a discriminant criterion in sparse unmixing approaches based on Matching Pursuit algorithms [64] to select the most suited endmembers. Only endmembers with "*a*" or "*b*" purity indicators, which refer to pure samples and samples with contaminant but without significant spectral effect, were considered.

Prior to the unmixing step, endmembers were resampled to match with AVIRIS spectral bands. Furthermore, image bands affected by atmosphere absorption were removed and replaced by interpolation between the last and next valid reflectance values. The same process was performed on the endmembers. In both the AVIRIS image and the endmember data sets, bands 106–114 and 152–168 were replaced.

IV. Model Evaluations

A. Assessment of Abundance Retrievals

The performances of the seven existing models presented in Section II, and the LMM and the proposed NBRU method were compared. The 94 spectra of crafted mineral mixtures were used in the process. For each existing model, the best set

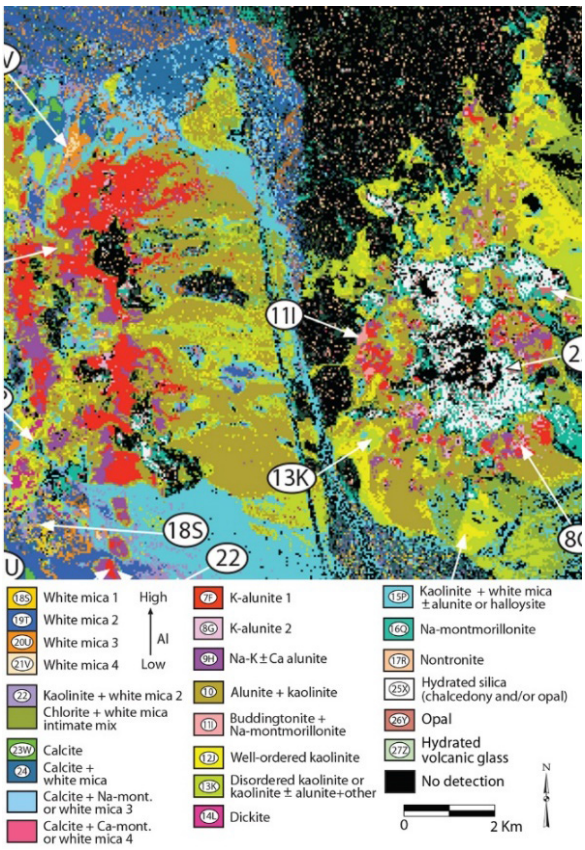


Fig. 4. Subset of the Tetracorder mineral map of spectrally dominant clays, micas, sulfates, and carbonates at Cuprite, modified from Swayze *et al.* [62]. Arrows indicate locations of samples collected by original authors.

of parameters was obtained by the minimization of the root-mean-squared error (RMSE) between measured and modeled reflectance spectra (21)

$$\mathrm{RMSE}(s, y) = \sqrt{\frac{1}{B}\|s - y\|_2^2} = \sqrt{\frac{1}{B}\sum_{i=1}^{B}(s_i - y_i)^2} \qquad (21)$$

where B is the number of spectral bands, s and y are the observed and modeled spectra, respectively. The set of abundances and extra-parameters that minimizes the cost function (21) is considered as optimum. The same approach was applied with NBRU using spectra of NBR (s^* and y^*) instead of reflectance.

In this experiment, the proportions of the minerals at the surface of the samples are assumed to be representative of their volume fraction. The latter is converted to mass abundances through their specific gravities [58], [59] or by inverting relationship (14) for Hapke's model. Performances were assessed by computing the mean abundance error between actual and retrieved mass abundances.

Parameter ranges were limited by setting upper and lower bounds as well as equality and inequality conditions in accordance with the requirements of the models. Only endmembers that were parts of the mixtures were used in the minimization processes.

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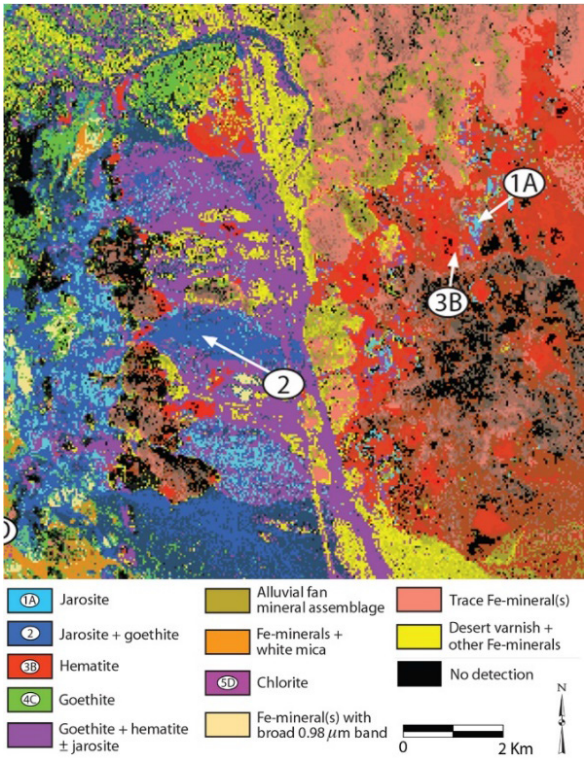


Fig. 5. Subset of the Tetracorder mineral map of spectrally dominant Fe-bearing minerals at Cuprite, modified from Swayze *et al.* [62]. Arrows indicate locations of samples collected by original authors.

B. Assessment of Mineral Spatial Distribution Mapping

The same unmixing strategy was applied on a subset of the AVIRIS image of Cuprite to assess the models' ability to map the spatial distributions of mineral species.

Swayze *et al.* [62] extensively studied the area combining published geologic maps, field observations, sample analysis, geophysical surveys, and mineral mapping using the Tetracorder System [65] on an AVIRIS image acquired on June 23, 1995. Authors produced two maps used in this article: one is the Fe-bearing minerals map and the other one is that of clays, micas, sulfates, and carbonates (Figs. 4 and 5). Both maps were used as validation data to assess the ability of the models to provide reliable spatial information on mineral distributions.

The two validation maps were projected to match with the subset of the AVIRIS image used in the unmixings. Projections were performed by fitting quadratic functions with common locations manually identified between the AVIRIS image and the two validation maps (Figs. 4 and 5). While a gap of two years separates the acquisition of the AVIRIS image used in [62] and the one used in this article, only minor changes were observed thanks to the desert inhabited nature of the area.

From the 14 minerals mentioned in the reference maps (Figs. 4 and 5), we limited our comparisons to the nine most dominant minerals with clear spatial patterns, namely, alunite, buddingtonite, calcite, dickite, hematite, goethite, jarosite, kaolinite, and opal.

Because of the important number of variables to retrieve, GBM, was not evaluated on the AVIRIS image. Indeed, this model requires all second-order interactions to be quantified independently, leading to retrieve 105 parameters with 14 endmembers. The required computing time to reach an optimal set of parameters with reasonable accuracy was considered a deal breaker.

Also, RT models are required to know the grain size of the samples at the origin of the endmembers. However, this characteristic is generally not available. Consequently, Hapke's and Shkuratov's models were not tested on the AVIRIS image. We, therefore, limited the comparison of mapping performances to the LMM, the FM, the PPNMM, the MLM, the MHPNMM, and the NBRU.

A two-step strategy was applied to assess the ability of the models to provide reliable spatial distributions.

- The class of the mineral considered was first extracted from the appropriate validation map (Fig. 4 or 5). This step provided binary maps used as validation (see Fig. 7). Because the class *Calcite* has particularly few pixels and is partly hidden by arrows and sample identifications, the multimineral class *Calcite + white micas* was also extracted.
- A binary classification was then performed on the nine abundance maps of the six models. For all maps, the applied thresholding value was the one that maximized the Kappa coefficient.

Kappa coefficients are presented in Section V-B. Binary classification maps for the five nonlinear approaches are also illustrated.

Although per-pixel endmember selection has been addressed in numerous unmixing studies [66]–[70], this experiment intends to evaluate how spectral mixing models handle the inaccuracy and variability of the endmembers in natural scenes. A single set of 14 endmembers has therefore been used to unmix the entire Cuprite image. It is worth noticing that methods have been proposed to alleviate the endmember variability problem (see Introduction). Again, according to the aim of this article, none of them has been considered here in order to evaluate the sensitivity of the mixing models.

V. Results and Discussion

A. Evaluation of Unmixing Models Using Mineral Mixtures

The unmixing methods were performed on 94 spectra of mineral mixtures. Performances were evaluated on differences between real and retrieved mass abundances expressed as mean abundance error. No endmember variability is expected in the data set since the mineral samples used in the endmember measurements are those mixed in the crafted multimineral mixtures. Consequently, performances of the nine unmixing methods in retrieving mineral abundances only rely on their ability to capture the nonlinear relations encountered in intimate mixtures. Histograms in Fig. 6 and Table I summarizes the results.

Histograms in Fig. 6 and Table I show the clear inability of the linear and bilinear models to capture the complex relationship between the abundance of a mineral and its spectral influence.

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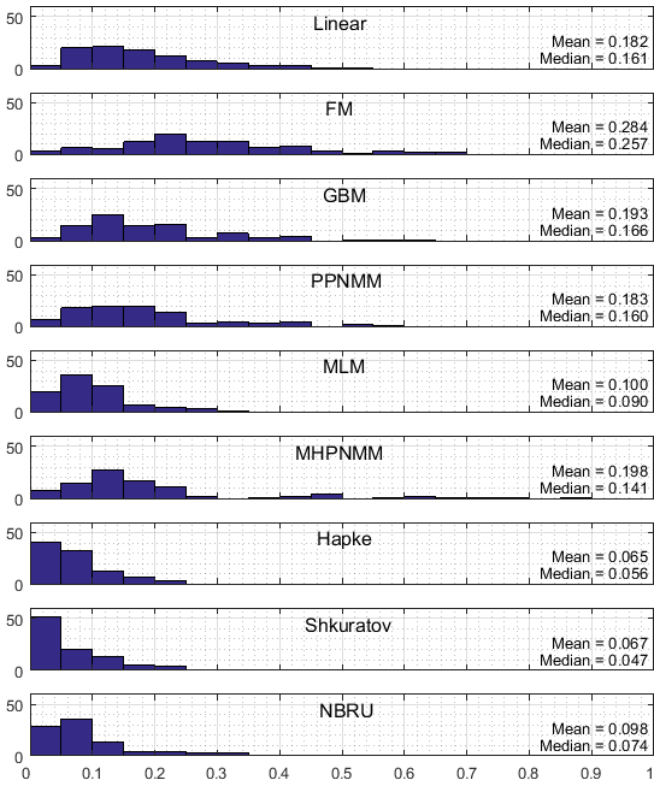


Fig. 6. Histograms of the unmixing performances, in terms of mean abundance error, for the 94 mixtures tested. Means and medians are added for information.

TABLE I
FREQUENCY OF BEST PERFORMANCE FOR THE 11 MODELS APPLIED ON THE SPECTRA OF THE 94 MINERAL MIXTURES

Model	Frequency of best estimation
LMM	2
FM	2
GBM	0
PPNMM	2
MLM	10
MHPNMM	7
Hapke	34
Shkuratov	21
NBRU	16

MLM obtained ten best estimations and mean and median errors of 10.7% and 10%, respectively. To better understand these contrasting performances with the bilinear models, the MLM has been tested without the transformation of reflectance into albedo spectra (10). This version of the model led to mean and median errors of 20.1% and 16.8%, respectively, which are of the same order of magnitude as the bilinear models. Note that these results are neither shown in Table I nor in Fig. 6 for clarity purposes.

MHPNMM's results showed important inconsistencies, leading to the most accurate estimations of seven mineral mixtures (Table I) but also showing errors between 85% and 90% (Fig. 6). It thus appears that the addition of harmonic relationships in the model, quantified by a free parameter b (11), does not help to capture the spectral behavior of intimate mixtures.

Hapke's RT model outperformed the other models in Table I with 34 most accurate estimations. However, Shkuratov's model led to better statistics in Fig. 6, obtaining mean and median errors of 6.7% and 4.7% (vs. 6.5% and 5.6% for Hapke's model), respectively. Errors higher than 20% occurred, however, with several mixtures (Fig. 6).

NBRU obtained the best results among non-RT models with 16 most accurate estimations and mean and median errors of 10.7% and 6.8%, respectively (Fig. 6). These performances confirm the good aptitude of the proposed approach to capture the nonlinear relationship between a mineral abundance and its spectral contribution. Table II hereunder illustrates the performances of NBRU for 11 mixtures.

The above observations confirm the inability of macroscopic models to handle the spectral mixtures of particulate surfaces. Integrating the logarithmic trend of the relationship between a mineral abundance and its spectral influences (Fig. 1) is thus a necessity to retrieve quantitative information with reasonable accuracy.

The behaviors of the nine methods vary largely with the mineral assemblage encountered. Three of them were particularly challenging, gathering the highest errors. Detailed results for these three mixtures are introduced in Tables III–V.

For both RT models, mixtures in Table III led to mean errors higher than 20%. A hypothesis to explain these deviations is the frequent fibrous or foliated aspect of chlorite minerals [58]. Indeed, both RT models assume a similar shape of the particle of the endmembers in the mixtures. The foliated aspect classically presented by muscovite grains [58] may also cause the errors shown in Table IV. Another source of error related to the mineral shapes may be the unrepresentativeness of the samples' surfaces with their actual volume fractions.

Finally, important errors can also be observed in the estimated abundances of mineral mixtures involving graphite (Table V). The latter are due to the highly absorbing nature of the mineral, which violates the limitations of the RT models [31], [32], [53]. The other models show higher errors with these three mineral assemblages too (Tables III–V).

Nevertheless, abundance results demonstrate the higher accuracy of RT models when facing intimate mixtures of minerals. The errors also pinpoint the sensitivity of unmixing models to the physical properties of the particles and to the importance of the assumptions made.

However, RT models are required to know complementary information on the endmembers used. Consequently, they must be excluded when only endmember reflectance spectra are available. To better compare non-RT results, Table VI hereunder shows how the linear and bilinear models are outperformed by the MLM and NBRU. The latter obtained particularly better results with 41 best estimations of the 94 mixtures tested.

B. Evaluation of Unmixing Models Using AVIRIS Hyperspectral Data

In this section, the Cuprite AVIRIS image was used to assess the mapping abilities of FM, PPNMM, MLM, MHPNMM,

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TABLE II

MEAN ERRORS OF THE CALCULATED ABUNDANCES FOR HYPERSTHENE-AUGITE-LABRADORITE MIXTURES. THE TABLE CONTAINS SPECTRA ACQUIRED BY TIM MCCOY WITH THE NASA RELAB FACILITY. SAMPLES IDS FROM RELAB CATALOG ARE ALSO PROVIDED FOR REPRODUCIBILITY

		AG-TJM-009	AG-TJM-010	AG-TJM-011	Mean Errors									
	ID Relab	Hypersthene	Augite	Labradorite	Linear	FM	GBM	PPNMM	MLM	MHPNMM	Hapke	Shkuratov	NBRU	Best
Mixtures	AG-TJM-027	0.45	0.1	0.45	0.084	0.196	0.084	0.075	0.036	0.054	0.030	0.038	0.025	NBRU
	AG-TJM-028	0.4	0.2	0.4	0.099	0.159	0.099	0.095	0.047	0.112	0.058	0.057	0.033	NBRU
	AG-TJM-029	0.35	0.3	0.35	0.119	0.215	0.120	0.119	0.070	0.116	0.074	0.069	0.024	NBRU
	AG-TJM-030	0.3	0.4	0.3	0.157	0.267	0.163	0.164	0.097	0.175	0.054	0.046	0.027	NBRU
	AG-TJM-031	0.25	0.5	0.25	0.168	0.300	0.181	0.185	0.100	0.187	0.062	0.050	0.027	NBRU
	AG-TJM-042	0.135	0.1	0.765	0.079	0.150	0.079	0.057	0.066	0.022	0.059	0.055	0.069	MHPNMM
	AG-TJM-043	0.12	0.2	0.68	0.123	0.190	0.123	0.108	0.101	0.066	0.046	0.035	0.026	NBRU
	AG-TJM-044	0.105	0.3	0.595	0.136	0.223	0.136	0.125	0.129	0.096	0.067	0.052	0.042	NBRU
	AG-TJM-045	0.09	0.4	0.51	0.165	0.268	0.165	0.159	0.147	0.146	0.067	0.048	0.035	NBRU
	AG-TJM-046	0.075	0.5	0.425	0.206	0.317	0.208	0.207	0.154	0.216	0.046	0.025	0.032	Shkuratov
	AG-TJM-051	0.2	0.6	0.2	0.192	0.340	0.206	0.212	0.132	0.195	0.045	0.033	0.038	Shkuratov

TABLE III

MEAN ERRORS OF THE CALCULATED ABUNDANCES FOR CALCITE-CHLORITE MIXTURES. THE TABLE CONTAINS SPECTRA ACQUIRED BY CARLE M. PIETERS WITH THE NASA RELAB FACILITY. SAMPLES IDS FROM RELAB CATALOG ARE ALSO PROVIDED FOR REPRODUCIBILITY

		XT-CMP-010	XT-CMP-014	Mean Errors									
	ID RELAB	Calcite	Chlorite	Linear	FM	GBM	PPNMM	MLM	MHPNMM	Hapke	Shkuratov	NBRU	Best
Mixtures	XT-CMP-011	0.75	0.25	0.460	0.559	0.559	0.547	0.218	0.704	0.181	0.215	0.308	Hapke
	XT-CMP-012	0.5	0.5	0.340	0.395	0.395	0.392	0.154	0.085	0.137	0.151	0.208	MHPNMM
	XT-CMP-013	0.25	0.75	0.213	0.222	0.222	0.229	0.125	0.240	0.166	0.130	0.151	MLM
	XT-CMP-018	0.9	0.1	0.411	0.563	0.545	0.546	0.192	0.634	0.124	0.160	0.248	Hapke
	XT-CMP-019	0.8	0.2	0.501	0.602	0.602	0.600	0.262	0.688	0.219	0.254	0.347	Hapke
	XT-CMP-020	0.5	0.5	0.378	0.417	0.417	0.433	0.187	0.490	0.207	0.212	0.285	MLM

TABLE IV

MEAN ERRORS OF THE CALCULATED ABUNDANCES FOR QUARTZ-ALUNITE AND QUARTZ-MUSCOVITE MIXTURES. THE TABLE CONTAINS SPECTRA ACQUIRED BY CARLE M. PIETERS WITH THE NASA RELAB FACILITY. SAMPLES IDS FROM RELAB CATALOG ARE ALSO PROVIDED FOR REPRODUCIBILITY

		XT-CMP-001	XT-CMP-005	XT-CMP-009	Mean Errors									
	ID Relab	Quartz	Alunite	Muscovite	LMM	FM	GBM	PPNMM	MLM	MHPNMM	Hapke	Shkuratov	NBRU	Best
Mixtures	XT-CMP-002	0.75	0.25	-	0.394	0.572	0.418	0.416	0.255	0.744	0.061	0.104	0.246	Hapke
	XT-CMP-003	0.5	0.5	-	0.276	0.383	0.320	0.311	0.103	0.041	0.013	0.022	0.191	Hapke
	XT-CMP-004	0.25	0.75	-	0.167	0.212	0.170	0.168	0.132	0.244	0.038	0.030	0.091	Shkuratov
	XT-CMP-006	0.75	-	0.25	0.405	0.685	0.421	0.420	0.274	0.434	0.182	0.184	0.372	Hapke
	XT-CMP-007	0.5	-	0.5	0.365	0.483	0.365	0.365	0.303	0.364	0.231	0.209	0.310	Shkuratov
	XT-CMP-008	0.25	-	0.75	0.227	0.253	0.233	0.232	0.189	0.283	0.181	0.136	0.205	Shkuratov

TABLE V

MEAN ERRORS OF THE CALCULATED ABUNDANCES FOR FORSTERITE-ENSTATITE AND FORSTERITE-ENSTATITE-GRAPHITE MIXTURES. THE TABLE CONTAINS SPECTRA ACQUIRED BY CARLE M. PIETERS WITH THE NASA RELAB FACILITY. SAMPLES IDS FROM RELAB CATALOG ARE ALSO PROVIDED FOR REPRODUCIBILITY

		JB-JLB-945-D	JB-JLB-A14-A	JB-JLB-A05	Mean Errors									
	ID Relab	Forsterite	Enstatite	Graphite	LMM	FM	GBM	PPNMM	MLM	MHPNMM	Hapke	Shkuratov	NBRU	Best
Mixtures	JB-JLB-A15	0.5	0.5	-	0.348	0.430	0.348	0.275	0.169	0.040	0.186	0.206	0.199	MHPNMM
	JB-JLB-A16	0.25	0.75	-	0.206	0.239	0.206	0.161	0.092	0.089	0.113	0.120	0.141	MHPNMM
	JB-JLB-A17	0.75	0.25	-	0.305	0.462	0.305	0.245	0.106	0.226	0.057	0.098	0.152	Hapke
	JB-JLB-A18	0.475	0.475	0.05	0.323	0.349	0.323	0.213	0.220	0.141	0.132	0.100	0.121	Shkuratov
	JB-JLB-A19	0.48	0.48	0.04	0.222	0.259	0.222	0.166	0.127	0.172	0.082	0.073	0.069	NBRU

and NBRU with remote sensing data. The criteria to select the models are given in Section IV-B.

To assess the performances of the six models, abundance maps were converted into two-class maps based on a thresholding abundance value. The threshold that showed the highest Kappa coefficient when compared with mineral maps of [62] (Figs. 4 and 5) was selected. The Kappa coefficients of the binary classifications are presented in Table VII.

Table VII shows that bilinear models obtained poor results, a notable exception being the buddingtonite and the calcite mapped by the FM. On the other hand, NBRU produced the best spatial distributions for seven of the nine minerals considered. For all methods, the mapping of goethite showed little agreement with the validation map (Fig. 5).

Fig. 7 hereunder shows binary classifications of abundance maps obtained by the five nonlinear approaches tested. The six minerals with the best global mapping results, namely

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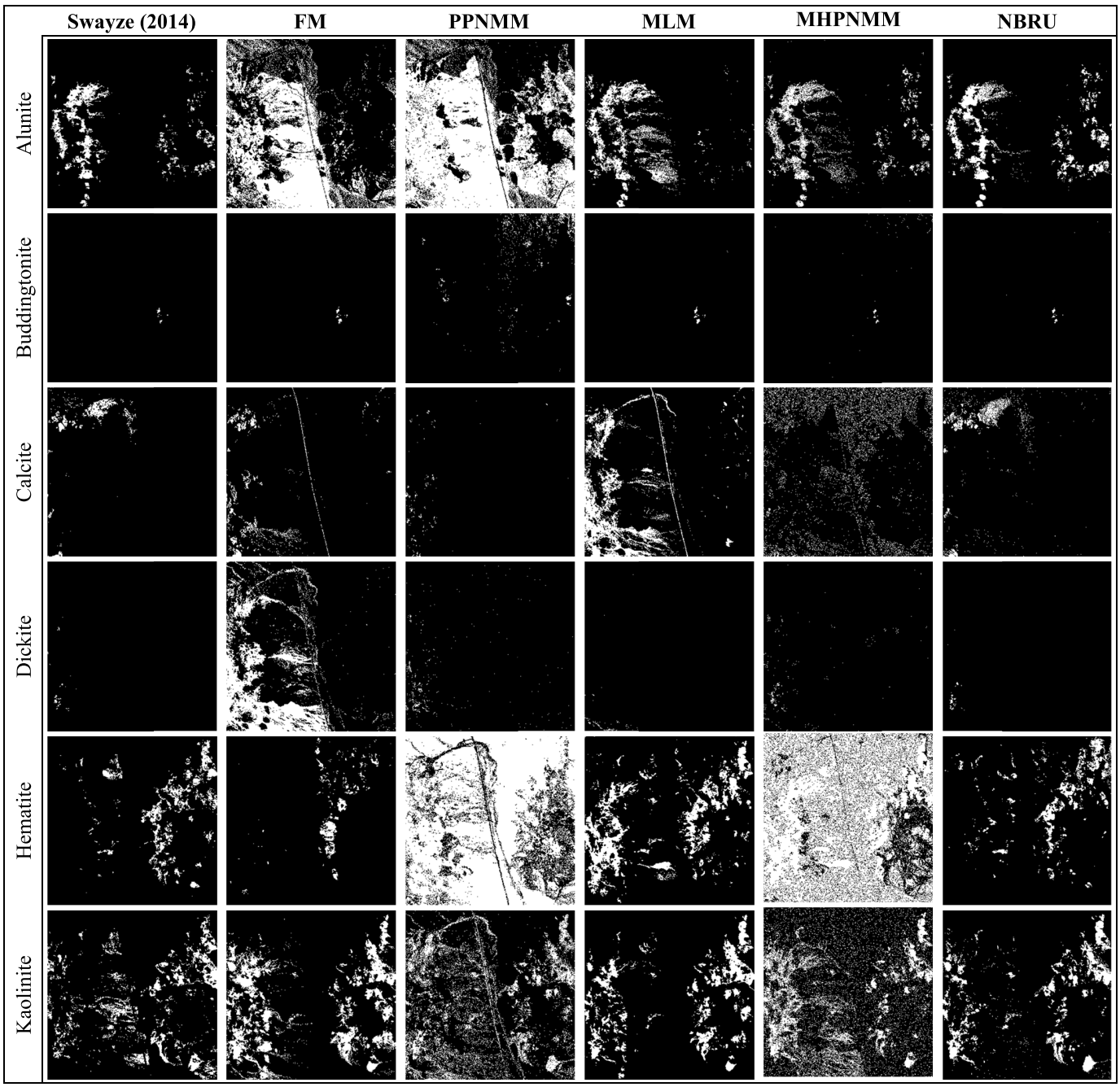


Fig. 7. Binary classification of abundance maps.

TABLE VI

FREQUENCY OF BEST PERFORMANCE FOR THE SEVEN NON-RT MODELS APPLIED ON THE SPECTRA OF THE 94 MINERAL MIXTURES

Model	Frequency of best estimation
LMM	2
FM	4
GBM	0
PPNMM	2
MLM	36
MHPNMM	9
NBRU	41

TABLE VII

KAPPA COEFFICIENTS OF THE BINARY CLASSIFICATIONS OF THE NINE MINERALS MAPPED

	LMM	FM	PPNMM	MLM	MHPNMM	NBRU
Alunite	0	0.018	0.074	0.439	0.529	**0.652**
Buddingtonite	0.147	0.618	0.009	0.640	0.429	**0.651**
Calcite	0	0.166	0.048	0.037	0.091	**0.469**
Dickite	0	0.006	0.087	0.047	0.234	**0.419**
Goethite	0.015	0.006	**0.050**	0.002	0.034	**0.050**
Hematite	0	0.024	0.007	0.243	0.029	**0.389**
Jarosite	0	0.140	0.056	**0.177**	0.040	0.067
Kaolinite	0.01	0.165	0.201	0.256	0.191	**0.402**
Opal	0.042	0.093	0.067	0.032	0.047	**0.187**

alunite, buddingtonite, calcite, dickite, hematite, and kaolinite, are illustrated. The mineral classes extracted from Figs. 4 and 5 are shown as well for comparison.

Obtained by FM, PPNMM, MLM, MHPNMM, and NBRU for six minerals. The patterns of the buddingtonite and the kaolinite map obtained by the FM and PPNMM show some

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similarities with the reference maps (first columns). Consistent spatial patterns of alunite were also retrieved by the MLM and the MHPNMM. The latter show, however, more speckles. Finally, NBRU obtained consistent spatial patterns for all six minerals illustrated in Fig. 7. These observations agree with the kappa coefficients shown in Table VII.

The similarities between the experimental results underline the necessity for the models to handle intimate spectral behaviors. However, none of the compared approaches provide reliable information for the spatial distribution of goethite. The mineral maps of jarosite and opal show poor consistencies with validation data as well. These results are interpreted as a consequence of an inappropriate selection of endmembers prior to the unmixing process. Both minerals show large spectral variabilities in terms of shape and amplitude, with important shifts of the absorption bands. Although the correlation criterion is an easy and computationally fast way to select endmembers, more elaborate methods may lead to different and eventually better results. Also, restraining the set of endmembers used in the unmixing to the only one present in the considered pixel can significantly improve the mapping results. Finally, combining the models with appropriate spectral preprocessing and using shape-based reconstruction metrics in the minimization process can help alleviating the errors due to endmember inaccuracies (see Section I).

VI. Conclusion

In this article, the ability of existing nonlinear models in retrieving mineral abundances and mapping their spatial distribution were tested. We also proposed a nonlinear unmixing approach called NBRU. The method is built on shape-based reasoning that assumes that the shape of a multimineral reflectance spectrum is the combination of the shape of its endmembers. Spectral shapes in the NBRU are expressed as neighbor-band ratios. Inverting the proposed approach consists in retrieving the combination of weights (or abundances) that minimizes the RMSE between the modeled and measured NBR spectra. This inversion strategy is similar to the classic LMM with reflectance spectra.

The advantages of NBRU are twofold. First, shape-based matching criteria have been proven to be more robust when dealing with endmember variabilities. These variabilities are frequently encountered in natural scenes because of illumination changes and BRDF-related effects. Second, using NBR allows a nonlinear relationship to be induced between the spectral contributions of the endmembers and their abundance. This nonlinear behavior expresses a well-observed higher influence of darker materials relative to lighter ones.

The performances of the models were tested on two hyperspectral data sets. The first was performed on 94 spectra of binary, ternary, and quaternary mineral mixtures selected in the NASA RELAB database. Performances of NBRU were compared to seven other nonlinear models and to the LMM. Since the endmembers used in the unmixing processes were those used in the crafted mixtures and because measurements were taken in labs with controlled illumination, no endmember variability is possible.

Results were dominated by the RT models with 34 best estimations for Hapke's model and 21 best estimations for Shkuratov's model. NBRU outperformed the remaining models with 16 mixtures while the MLM obtained the best results for 10 of them. However, RT models require complementary information on the endmembers used in the unmixing. When the latter are lacking, only non-RT models can be considered. Comparisons of non-RT results showed that MLM and NBRU led to significantly better estimations. Particularly, the latter reached 41 best abundance estimations of the 94 RELAB mixtures tested.

Accuracy of the models in calculating mineral abundances appears to largely depend on the mineral species encountered. Particularly, mineral shapes have a key role in the spectral behavior of the involved endmembers and significantly affect the performances of all unmixing models.

The second experiment consisted in a direct application of the unmixing models on remote sensing data. The data set used is a hyperspectral image of the well-studied Cuprite area. Again, MLM and NBRU obtained better results with several minerals, showing spatial patterns similar to the validation maps and/or better kappa coefficients. NBRU performed particularly well with the highest kappa coefficients for seven of the nine mapped minerals. Thus, the proposed shape-based approach appears to better handle the endmember variability encountered when unmixing natural scenes and when *in situ* data are lacking.

In conclusion, the poor performances of the linear and bilinear models underline their inability in mapping mineral intimate mixtures without additional preprocessing steps. Pixel-specific endmembers and alternative criteria in the minimization process may lead to significant improvement. Consequently, bilinear models should not be used alone as reference to validate new approaches neither to simulate intimate spectral mixtures. Instead, only models specifically developed for this purpose should be investigated. Thanks to its best results among non-RT models in both abundance estimations and mapping performances, the proposed NBRU approach could be considered when RT models are not applicable.

Future work includes the applicability of NBRU in other contexts with important multiscattering, like vegetation or planetary regolith. Model performances with more elaborate endmembers selections and spectral preprocessing methods are also worth being investigated.

Acknowledgment

The authors would like to thank Jet Propulsion Laboratory (JPL) for making the AVIRIS hyperspectral reflectance image of Cuprite publicly available, the Reflectance Experiment Laboratory (RELAB) from Brown University for the free access to their spectral database, and the United States Geological Survey (USGS) also for the free access to the Spectral Library Version 7. They would also like to thank the anonymous reviewers for their suggestions and insightful questions that helped to improve the clarity and scientific quality of this article.

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